

4.Design Zero-shot, One-shot, and Few-shot prompts to generate a professional email requesting leave due to illness. Compare the generated emails based on tone, grammar, formatting, and completeness.

ZERO-SHOT:

```
from transformers import pipeline
```

```
generator = pipeline("text-generation", model="gpt2")
```

```
prompt = ""
```

Write a professional email requesting two days of leave due to illness.

```
""
```

```
result = generator(
```

```
    prompt,
```

```
    max_length=180,
```

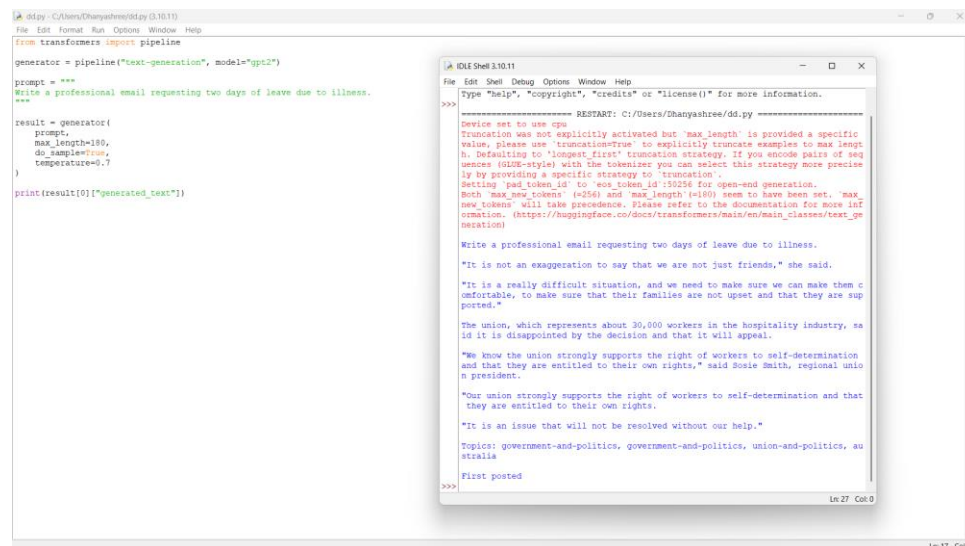
```
    do_sample=True,
```

```
    temperature=0.7
```

```
)
```

```
print(result[0]["generated_text"])
```

output



The screenshot shows a Jupyter Notebook on the left and a terminal window on the right. The Jupyter Notebook contains the following code:

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = ""
Write a professional email requesting two days of leave due to illness.

result = generator(
    prompt,
    max_length=180,
    do_sample=True,
    temperature=0.7
)

print(result[0]["generated_text"])
```

The terminal window on the right shows the output of the code. It starts with a message about the device being set to use CPU. Then, it displays the generated text:

```
Write a professional email requesting two days of leave due to illness.

It is not an exaggeration to say that we are not just friends," she said.

It is a really difficult situation, and we need to make sure we can make them c
omfortable, to make sure that their families are not upset and that they are sup
ported."

The union, which represents about 30,000 workers in the hospitality industry, sa
id it is disappointed by the decision and that it will appeal.

"We know the union strongly supports the right of workers to self-determination
and that they are entitled to their own rights," said Sosie Smith, regional unio
n president.

"Our union strongly supports the right of workers to self-determination and that
they are entitled to their own rights.

It is an issue that will not be resolved without our help."

Topics: government-and-politics, government-and-politics, union-and-politics, au
stralia

First posted
```

ONE-SHOT

```
from transformers import pipeline
```

```
generator = pipeline("text-generation", model="gpt2")
```

```
prompt = ""
```

Example

Subject: Leave Request

Dear Manager,

I request one day of leave due to a family emergency. I will resume work tomorrow.

Thank you.

Regards,

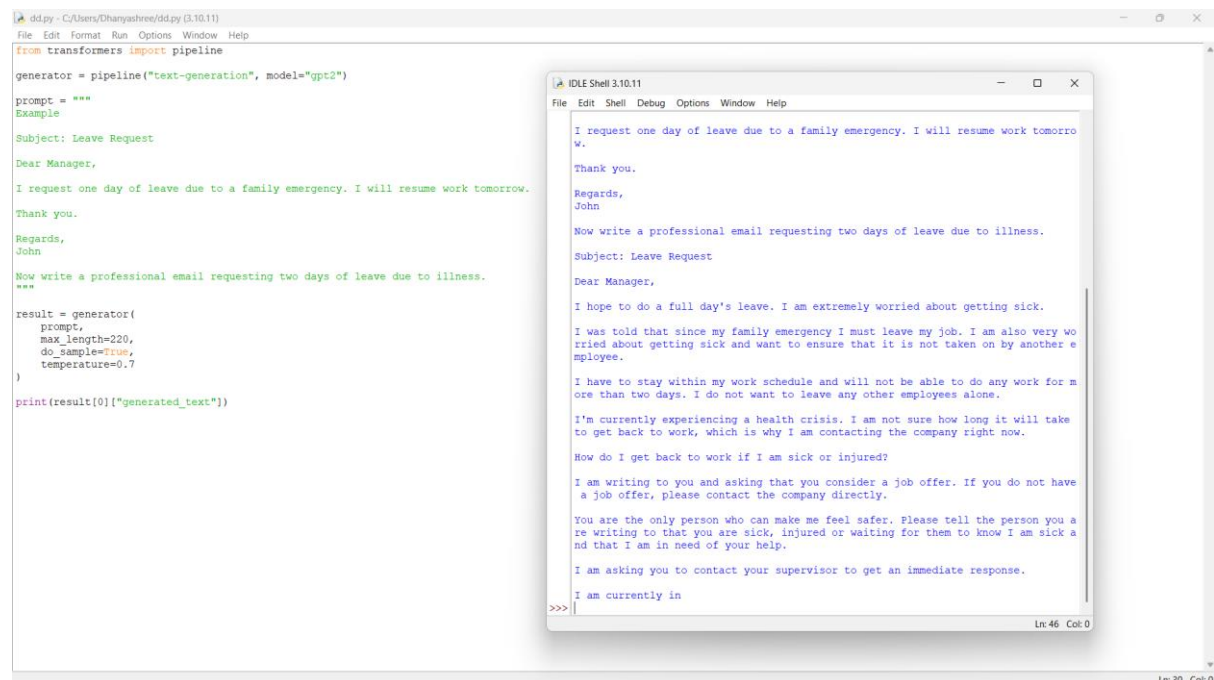
John

Now write a professional email requesting two days of leave due to illness.

=====

```
result = generator(  
    prompt,  
    max_length=220,  
    do_sample=True,  
    temperature=0.7  
)  
  
print(result[0]["generated_text"])
```

output:



The image shows a screenshot of a Python script in an IDE (dd.py) and its output in an IDLE Shell. The script uses the GPT-2 model to generate a professional email requesting two days of leave due to illness. The output is a detailed email body.

```
dd.py - C:/Users/Dhanyashree/dd.py (3.10.11)  
File Edit Format Run Options Window Help  
from transformers import pipeline  
  
generator = pipeline("text-generation", model="gpt2")  
  
prompt = """  
Example  
Subject: Leave Request  
Dear Manager,  
I request one day of leave due to a family emergency. I will resume work tomorrow.  
Thank you.  
Regards,  
John  
Now write a professional email requesting two days of leave due to illness.  
"""  
  
result = generator(  
    prompt,  
    max_length=220,  
    do_sample=True,  
    temperature=0.7  
)  
  
print(result[0]["generated_text"])
```

IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
I request one day of leave due to a family emergency. I will resume work tomorrow.
Thank you.
Regards,
John
Now write a professional email requesting two days of leave due to illness.
Subject: Leave Request
Dear Manager,
I hope to do a full day's leave. I am extremely worried about getting sick.
I was told that since my family emergency I must leave my job. I am also very worried about getting sick and want to ensure that it is not taken on by another employee.
I have to stay within my work schedule and will not be able to do any work for more than two days. I do not want to leave any other employees alone.
I'm currently experiencing a health crisis. I am not sure how long it will take to get back to work, which is why I am contacting the company right now.
How do I get back to work if I am sick or injured?
I am writing to you and asking that you consider a job offer. If you do not have a job offer, please contact the company directly.
You are the only person who can make me feel safer. Please tell the person you are writing to that you are sick, injured or waiting for them to know I am sick and that I am in need of your help.
I am asking you to contact your supervisor to get an immediate response.
I am currently in
>>> Ln: 46 Col: 0

FEW-SHOT:

```
from transformers import pipeline
```

```
generator = pipeline("text-generation", model="gpt2")
```

```
prompt = ""
```

Example 1

Subject: Casual Leave

Dear Sir,

I request one day of casual leave due to personal work.

Regards,

Rahul

Example 2

Subject: Medical Leave

Dear Manager,

I am suffering from a fever and request two days of medical leave.

Regards,

Anjali

Now write a professional email requesting two days of leave due to illness.

```
"""
```

```
result = generator(
```

```
    prompt,
```

```
    max_length=250,
```

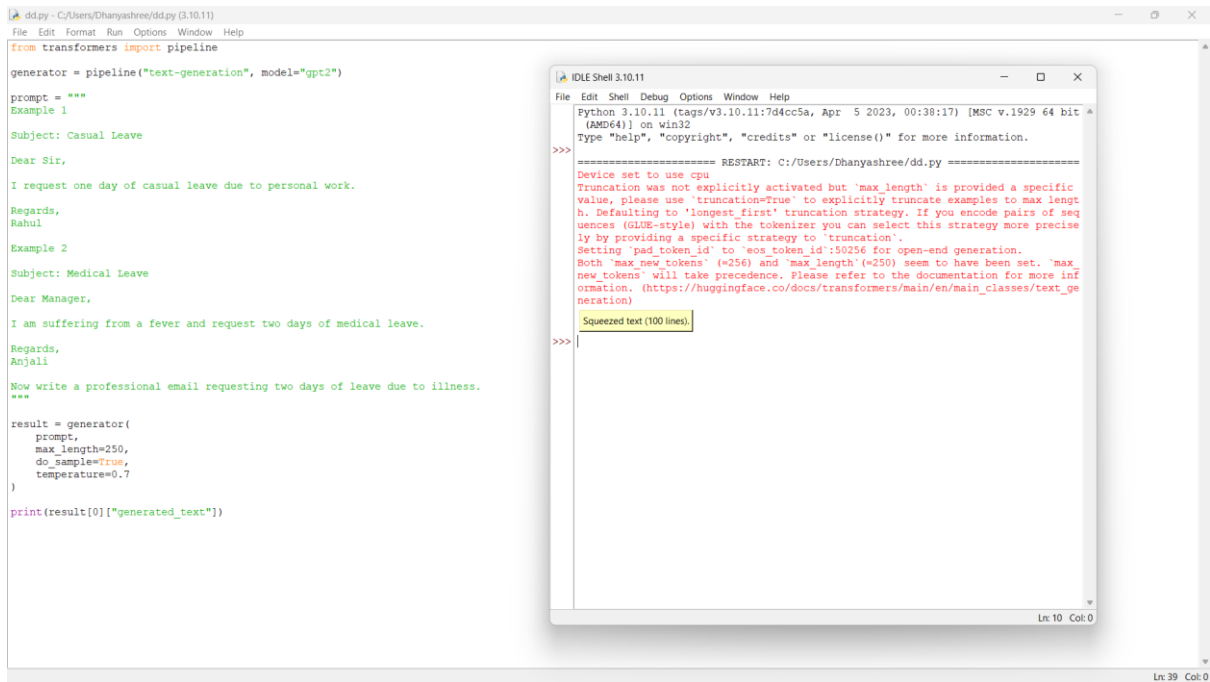
```
    do_sample=True,
```

```
    temperature=0.7
```

```
)
```

```
print(result[0]["generated_text"])
```

output:



The image shows a screenshot of a code editor window titled 'dd.py - C:/Users/Dhanyashree/dd.py (3.10.11)'. The code defines a text-generation pipeline using GPT-2. It includes two examples of prompts and the corresponding generated text. The first example prompt is 'Subject: Casual Leave' and the generated text is 'Dear Sir, I request one day of casual leave due to personal work. Regards, Rahul'. The second example prompt is 'Subject: Medical Leave' and the generated text is 'Dear Manager, I am suffering from a fever and request two days of medical leave. Regards, Anjali'. The code then uses a generator to produce text based on a prompt, with parameters: max_length=250, do_sample=True, and temperature=0.7. The output of the generator is printed as 'generated_text'.

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = """
Example 1
Subject: Casual Leave

Dear Sir,

I request one day of casual leave due to personal work.

Regards,
Rahul

Example 2
Subject: Medical Leave

Dear Manager,

I am suffering from a fever and request two days of medical leave.

Regards,
Anjali

Now write a professional email requesting two days of leave due to illness.
"""

result = generator(
    prompt,
    max_length=250,
    do_sample=True,
    temperature=0.7
)

print(result[0]["generated_text"])
```

The output window shows the execution of the script. It displays the Python version (3.10.11), the date and time (Apr 5 2023, 00:38:17), and the device (AMD64). It also shows the command 'Type "help", "copyright", "credits" or "license()" for more information.' and the command 'RESTART: C:/Users/Dhanyashree/dd.py'. The output window is currently empty, indicating that the script has not yet executed.